

Unit 30: Inverse Functions and Their Derivatives

1. The idea, in plain words

A function takes an input and produces an output. An inverse function undoes that — it takes the output and hands you back the original input. If f turns 3 into 7, then f^{-1} turns 7 back into 3.

A crucial warning right up front: $f^{-1}(x)$ does NOT mean $\frac{1}{f(x)}$. This is probably the single most common misconception with inverse functions — the “ -1 ” here is not an exponent. It’s just notation meaning “the inverse function,” a completely different function from f itself.

Picture it like undoing a set of instructions. If f says “double the number, then add 3,” the inverse has to undo those steps in reverse order: “subtract 3, then divide by 2.” Finding an inverse algebraically works the exact same way: write $y = f(x)$, then swap the roles of x and y (since input and output are trading places), and solve for y again. Whatever you get is $f^{-1}(x)$.

Not every function has an inverse. A function only has an inverse if it’s one-to-one — meaning every output comes from exactly one input (no two different x -values ever produce the same y -value). A function that’s always increasing, or always decreasing, across its whole domain automatically qualifies (it can’t ever repeat an output).

The real payoff of this unit — the Inverse Function Derivative formula. Sometimes finding an explicit formula for $f^{-1}(x)$ is difficult or flat-out impossible (imagine trying to solve a complicated cubic or a fifth-degree polynomial for x in terms of y). The wonderful news: you can still find the derivative of the inverse at a specific point, without ever writing down the inverse function itself — just using facts about the original function f .

2. Toolbox

Finding an inverse algebraically:

1. Write $y = f(x)$.
2. Swap x and y .
3. Solve the new equation for y .
4. That result is $f^{-1}(x)$.

Verifying two functions are inverses of each other:

$$f(f^{-1}(x)) = x \quad \text{and} \quad f^{-1}(f(x)) = x$$

One-to-one requirement: f has an inverse only if it’s one-to-one. A function that is strictly increasing ($f'(x) > 0$ everywhere) or strictly decreasing ($f'(x) < 0$ everywhere) on its domain is automatically one-to-one.

The Inverse Function Derivative formula — the star of this unit:

$$(f^{-1})'(a) = \frac{1}{f'(f^{-1}(a))}$$

In plain words: to find the derivative of the inverse at the point a , first figure out which input of f produces the output a (that input is $f^{-1}(a)$), then take the reciprocal of f' evaluated at that input.

A convenient way to organize this when you're given $f(p) = a$ directly: since $f^{-1}(a) = p$,

$$(f^{-1})'(a) = \frac{1}{f'(p)}$$

3. Common mistakes

- Confusing $f^{-1}(x)$ with $\frac{1}{f(x)}$. These are completely different things — always keep this distinction sharp.
- Forgetting to swap x and y before solving, when finding an inverse explicitly — solving for y without swapping just gives you back the original function.
- Applying the derivative formula with the wrong input. The formula needs f' evaluated at $f^{-1}(a)$ — that is, at whatever input of f produces the output a — not at a itself.
- Forgetting to first identify $f^{-1}(a)$ before plugging into the formula. If you're given that $f(p) = a$, then $f^{-1}(a) = p$ — use that p -value inside $f'(\cdot)$.
- Assuming every function has an inverse. Always check (or be told) that f is one-to-one before treating f^{-1} as a legitimate function.

4. Problem Set

Warm-up

1. Find $f^{-1}(x)$ for $f(x) = 2x + 3$.
2. Find $f^{-1}(x)$ for $f(x) = 5x - 2$.
3. Find $f^{-1}(x)$ for $f(x) = 4x$.
4. Find $f^{-1}(x)$ for $f(x) = \frac{x}{2} + 1$.
5. Verify that $f(x) = 3x - 6$ and $g(x) = \frac{x+6}{3}$ are inverses of each other by checking that $f(g(x)) = x$.
6. Find $f^{-1}(x)$ for $f(x) = -2x + 7$.

Standard

7. Find $f^{-1}(x)$ for $f(x) = x^3$.
8. Find $f^{-1}(x)$ for $f(x) = \sqrt{x-2}$, $x \geq 2$, and state the domain of the inverse.
9. Find $f^{-1}(x)$ for $f(x) = x^2 + 3$, $x \geq 0$.
10. Given $f(x) = 2x + 1$: (a) find $f^{-1}(x)$ explicitly and differentiate it directly; (b) use the Inverse Function Derivative formula instead. Confirm both methods agree.
11. Given $f(x) = x^3$, use the Inverse Function Derivative formula to find $(f^{-1})'(8)$ — without finding a general formula for $f^{-1}(x)$.
12. Given $f(x) = x^2 + 1$ for $x \geq 0$, with $f(2) = 5$, use the Inverse Function Derivative formula to find $(f^{-1})'(5)$.

Challenge

13. Given $f(x) = 6x - 1$, find the inverse f^{-1} of $f(x)$.
14. Let $f(x) = x^3 + x^2 + 6x + 1$, $x \geq 0$. Find the value of $\frac{d(f^{-1})}{dx}$ at the point $a = 1 = f(0)$.

15. Given $f(x) = x^5 + 2x + 1$, explain why finding $f^{-1}(x)$ explicitly would be extremely difficult. Then, using the fact that $f(1) = 4$, use the Inverse Function Derivative formula to find $(f^{-1})'(4)$.
16. Given $f(x) = x^3 + x$: first verify that f is one-to-one, then find $(f^{-1})'(2)$, given that $f(1) = 2$.
17. Given $f(x) = 2x^5 + x^3 + 1$, and knowing $f(1) = 4$, find $(f^{-1})'(4)$.

Applied

18. A temperature conversion function $f(C) = 1.8C + 32$ converts Celsius to Fahrenheit. Find $f^{-1}(F)$ (converting Fahrenheit back to Celsius), and use it to convert $98.6^\circ F$ to Celsius.
19. A company's cost to produce x units is $C(x) = 50x + 200$ dollars. Find the inverse function (giving the number of units producible for a given budget), and use it to find how many units can be produced with a budget of \$1200.
20. A demand function relating price to quantity is $p(q) = 100 - 2q$. Find the inverse function $q(p)$, and briefly interpret what it represents.
21. A rocket's height (in meters) as a function of fuel burned (in tons) is modeled by $h(x) = x^3 + 2x$ for $x \geq 0$, with $h(2) = 12$. Use the Inverse Function Derivative formula to find $(h^{-1})'(12)$, and interpret what this value represents physically.